

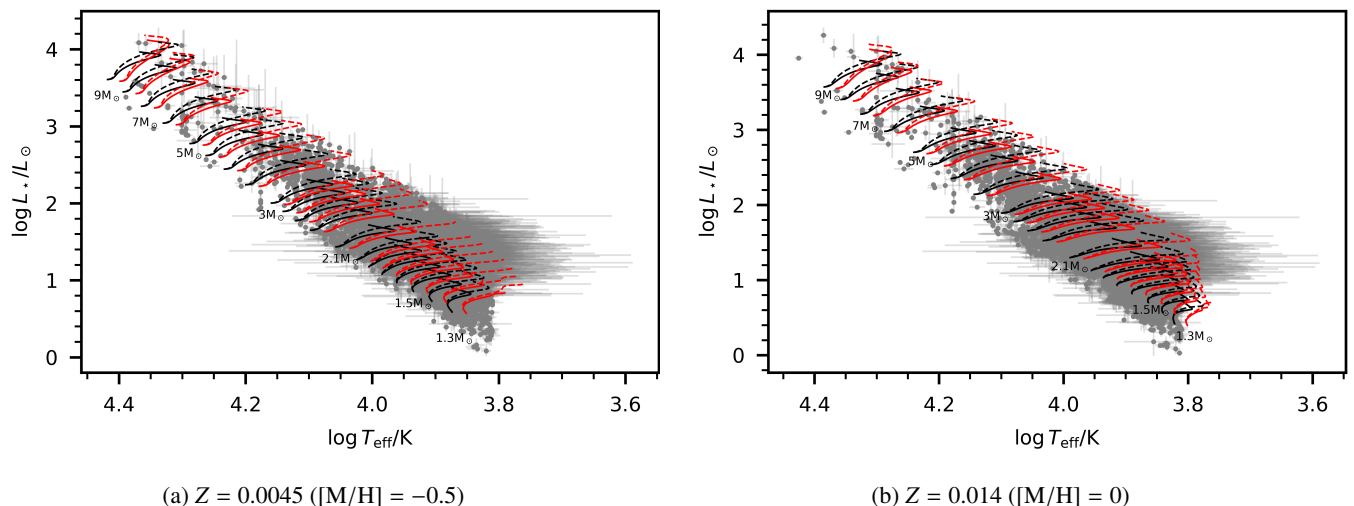


Fig. 2: Hertzsprung-Russell diagrams showing the sample of g-mode pulsators with $P_{\text{pred}} \geq 0.5$ (grey symbols with error bars). The evolutionary tracks show the extend of the grid of rotating stellar models, covering a mass range of $[1.3, 9] M_{\odot}$, where the black (red) colour indicates $\omega_0 = 0.05$ (0.55), and the solid (dashed) line style $f_{\text{CBM}} = 0.005$ (0.025).

is achieved with high-resolution spectroscopy (Gebruers et al. 2021), namely a median uncertainty of 108 K. The average relative uncertainty as a function of the effective temperature itself is constant, while for $\log L/L_{\odot}$, there is a decrease from ~ 8 per cent to ~ 3 per cent between $1 < \log L/L_{\odot} < 2$. We coupled these observables to two grids of stellar evolution models for rotating stars. These models have been constructed specifically for our purpose, by relying on input physics calibrated by the genuine *Kepler* γ Dor prograde dipole pulsators from asteroseismic modelling by Mombarg (2023). These models are compliant with the observed angular momentum of the stellar interior as probed by gravito-inertial asteroseismology following the method in Aerts et al. (2018) and summarised in Aerts (2021).

3. Grid-based parameter estimation

Our parameter estimation relies on the effective temperature and luminosity from *Gaia* DR3 as the two observables. These were coupled to the theoretically predicted values as derived from two grids of stellar evolution models with input physics calibrated by gravito-inertial asteroseismology. We first describe these model grids and then the method of the parameter estimation. Our computational setup, computed models, and results discussed below are publicly available on Zenodo¹.

3.1. Stellar structure and evolution models

We compute a grid of stellar structure and evolution models with the code MESA r24.03.1 (Paxton et al. 2011, 2013, 2015, 2018, 2019; Jermyn et al. 2023) covering masses, $M_{\star}$, between 1.3 and 9 $M_{\odot}$. We assume an exponentially decaying convective-boundary mixing (CBM, or overshoot) zone outside the convective core, where the radiative temperature gradient is taken and for which the slope of the diffusion coefficient with respect to the radial coordinate is set by a factor f_{CBM} times the local pressure scale height. We varied this parameter for each stellar mass between 0.005 and 0.025, based on the inferred distributions from the sample of 539 genuine *Kepler* γ Dor stars by Mombarg et al.

(2024). The switch from convective mixing to CBM was set to a distance of 0.005 times the local pressure scale height into the convection zone (overshoot_f0 in MESA). We also varied the initial fraction of Keplerian critical rotation at the zero-age main sequence (ZAMS), ω_0 , (assuming a uniform rotation profile at the ZAMS) between 5 and 55 per cent, following Mombarg et al. (2024). In MESA, the stellar structure equations are modified when rotation is enabled by redefining the radius coordinate, r , such that a sphere with this radius has the same volume as the distorted surface in the Roche model. This assumes that the angular velocity, Ω , is constant over isobars.

The chemical diffusion coefficient in the radiative envelope was computed from the rotational mixing predicted by the theoretical works of Zahn (1992) and Chaboyer & Zahn (1992),

$$D_{\text{rot}}(r > r_{\text{cc}}) = K \left(\frac{r}{N} \frac{d\Omega}{dr} \right)^2, \quad (1)$$

where K is the thermal diffusivity, N the Brunt-Väisälä frequency, and r_{cc} the radius of the convective core. The transport of angular momentum is done in a fully diffusive implementation, where we assumed a constant viscosity equal to $10^7 \text{ cm}^2 \text{ s}^{-1}$. This is consistent with the range derived for slowly rotating γ Dor stars near the terminal-age main sequence (TAMS) (Mombarg 2023). The evolution of a model is stopped if critical rotation is reached. This happens for a handful of models only.

We note that it is currently not well known which processes drive chemical mixing inside the radiative envelopes of main-sequence stars with a convective core. This is why we mimicked the unknown internal mixing processes and their effect on the luminosity and the effective temperature by encompassing the CBM parameter (f_{CBM}). Furthermore, we did not include microscopic diffusion, but note that this type of mixing alters the local Rosseland mean opacity. We compute this opacity for a solar mixture according to Asplund et al. (2009) using the data from the Opacity Project (Seaton 2005).

Adopting the present-day cosmic abundances of nearby early B-type stars from Nieva & Przybilla (2012) as initial composition, we compute a first grid of models for a metallicity $Z = 0.014$. This places many of the stars below the ZAMS line (Fig. 2, right panel). While the error bars based on the *Gaia* data

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Table 1: Parameter range covered by the grids used to train the CNF.

Parameter	Lower boundary	Upper boundary	Step size
$M_\star$	1.3 $M_\odot$	9 $M_\odot$	0.1-1 $M_\odot$
f_{CBM}	0.005	0.025	0.005
ω_0	0.05	0.55	0.05

are rather uncertain, this finding may represent shortcomings in the physics during the contraction phase until the ZAMS (we refer to [Zwintz & Steindl 2022](#) for an extensive discussion). However, the position of the stars below the ZAMS for solar metallicity is also in line with the sub-solar metallicity found consistently from both high-resolution ground-based spectroscopy and *Gaia* spectroscopy of γ Dor stars obtained by [Gebruers et al. \(2021\)](#) and [de Laverny et al. \(2024\)](#), respectively. For this reason, we also compute a second grid of models, for a metallicity $Z = 0.0045$ ($[M/H] = -0.5$) found as lower limit for a large sample of γ Dor stars by [de Laverny et al. \(2024\)](#) from *Gaia* spectroscopy. This second grid gives better agreement with the *Gaia* data as most stars are now well positioned above the ZAMS line (Fig. 2, left panel). For the grid with $Z = 0.014$, an initial helium mass fraction of $Y = 0.2612$ was used, and for the $Z = 0.0045$ grid $Y = 0.2495$ was used, following the chemical enrichment rate derived by [Verma et al. \(2019\)](#). The parameter ranges covered by the grids are listed in Table 1. We limited ourselves to a lower limit of 1.3 $M_\odot$, since none of the g-mode pulsators known to date has a mass below this value. Moreover, lower-mass models require a different treatment of rotation as magnetic breaking becomes important.

Convection is treated following mixing length theory ([Cox & Giuli 1968](#)), where we use the solar-calibrated value $\alpha_{\text{MLT}} = 1.8$ ([Choi et al. 2016](#)). Since stars in the mass range considered here do not have a significant convective envelope during the main sequence, and we rely on models calibrated from gravity modes probing the deep interior, our work does not depend on the exact choice of α_{MLT} . Furthermore, an Eddington grey atmosphere is assumed, with an opacity that is consistent with the local temperature and pressure of the atmosphere.

3.2. Interpolation with conditional normalising flows

We seek to interpolate the computed stellar evolution models in the three aforementioned physical quantities, $M_\star$, f_{CBM} , and ω_0 . The potential of conditional normalising flows (CNFs) to emulate grids of stellar models has been pointed out by [Hon et al. \(2024\)](#), to which we refer for details on the methodology. In brief, normalising flows represent a class of probabilistic machine learning tools with the capacity to make transformations between simple and more complex distributions. Here, we take a similar approach as presented by [Hon et al. \(2024\)](#), making use of the Zuko software package ([Rozet et al. 2023](#)).

The general principle of normalising flows is to train a neural network to learn the transformation for a simple probability distribution of a random variable z to a more complex distribution of a variable $y = f(z)$. The transformation can be conditioned based on labels c such that probability densities $p(y|c)$ can be estimated. The labels c (conditional variables) are in this case $M_\star$, f_{CBM} , and ω_0 . The output variables are the effective temperature (T_{eff} , in log scale), the luminosity ($L_\star/L_\odot$, in log scale), the rotation rate measured by the g modes at the convective core boundary (Ω_{gc}), the hydrogen mass fraction in the core compared to the initial one (X_c/X_{ini}), the mass of the convective core

($m_{\text{cc}}/M_\odot$), and the stellar radius ($R_\star/R_\odot$, in log scale). The CNF is trained on a data set of ~155,000 points representing the evolution models in each grid.

We draw 2.5 million samples from the trained CNF within the ranges of $M_\star$, f_{CBM} , and ω_0 covered by the grid of stellar models. The combination of these three conditional variables are sampled uniformly. To avoid oversampling near the ZAMS and TAMS, we resample from these 2.5 million points such that we are left with a more or less uniform initial distribution of X_c/X_{ini} values. This leaves us with at least one million samples. For each star in the sample, the likelihood of each sample drawn from the CNF is computed,

$$\mathcal{L} = \prod_{y \in \mathbf{Y}} \frac{1}{\sqrt{2\pi}\sigma_y} \exp\left(-\frac{(y_{\text{CNF}} - y_{\text{obs}})^2}{2\sigma_y^2}\right), \quad (2)$$

where σ_y denotes the observational uncertainty, and $\mathbf{Y} = \{\log T_{\text{eff}}, \log L/L_\odot\}$. We then compute weighted kernel density estimations (KDE) for $M_\star$, X_c/X_{ini} , $m_{\text{cc}}/M_\star$, and $\log R_\star/R_\odot$, where the likelihood values are taken as the weights. The optimal KDE bandwidth is determined using the Silverman method ([Silverman 1986](#)). The uncertainties are defined as the 68 per cent confidence interval. We discard stars that clearly fall outside the grid by imposing cuts at $3.81 < \log T_{\text{eff}} < 4.43$ and $\log L/L_\odot < 5.2 \log T_{\text{eff}} - 18.3$.

4. Derived evolutionary stellar parameters

Figure 3 shows the resulting KDEs of the maximum likelihood estimates (MLEs) of the inferred masses, the evolutionary stages (X_c/X_{ini}) as a proxy for the ages, the convective core masses, and the radii. We show the results for the three different probability thresholds of the classification results in [Hey & Aerts \(2024\)](#). We leave out stars that fall in either the lowest or the highest mass bin, as we consider them to be located outside the grid. Combined with the cuts mentioned above, this omits 748 stars for $Z = 0.0045$ and 373 stars for $Z = 0.014$ ($P_{\text{pred}} \geq 0.5$).

Two-sample Kolomogorov-Smirnov tests are performed to determine whether a different threshold in P_{pred} affects the underlying distributions of the inferred stellar parameters. We adopt as null hypothesis that the underlying distributions of both samples are equal and reject it for p values lower than 0.05. We find that the distributions for $M_\star$, X_c/X_{ini} , $m_{\text{cc}}/M_\star$ are not drawn for the same underlying distribution for the samples based on $P_{\text{pred}} = 0.2$ and $P_{\text{pred}} = 0.5$ (see Table 2). Yet, for the distribution of the radius, the null hypothesis cannot be rejected. When we omit stars above 4 $M_\odot$, as higher masses are under-sampled, the null hypothesis can also not be rejected for the distribution of X_c/X_{ini} . Furthermore, two-sample Kolomogorov-Smirnov tests for the distributions of $P_{\text{pred}} = 0.5$ and $P_{\text{pred}} = 0.8$ yield p values lower than 0.05 for all four inferred stellar parameters. These conclusions hold for both $Z = 0.0045$ and $Z = 0.014$. We do caution over-interpretation of these tests because the sample sizes are quite different (Fig. 1).

We estimate the uncertainty of the derived distributions by means of bootstrapping, where each measurement is perturbed independently within its uncertainty interval following a uniform probability². The results of 1000 iterations are shown in

² Note that the resulting distributions of the MLEs do not necessarily follow a normal distribution. Therefore, we take the more conservative assumption to sample from a uniform distribution within the 68%-confidence interval.

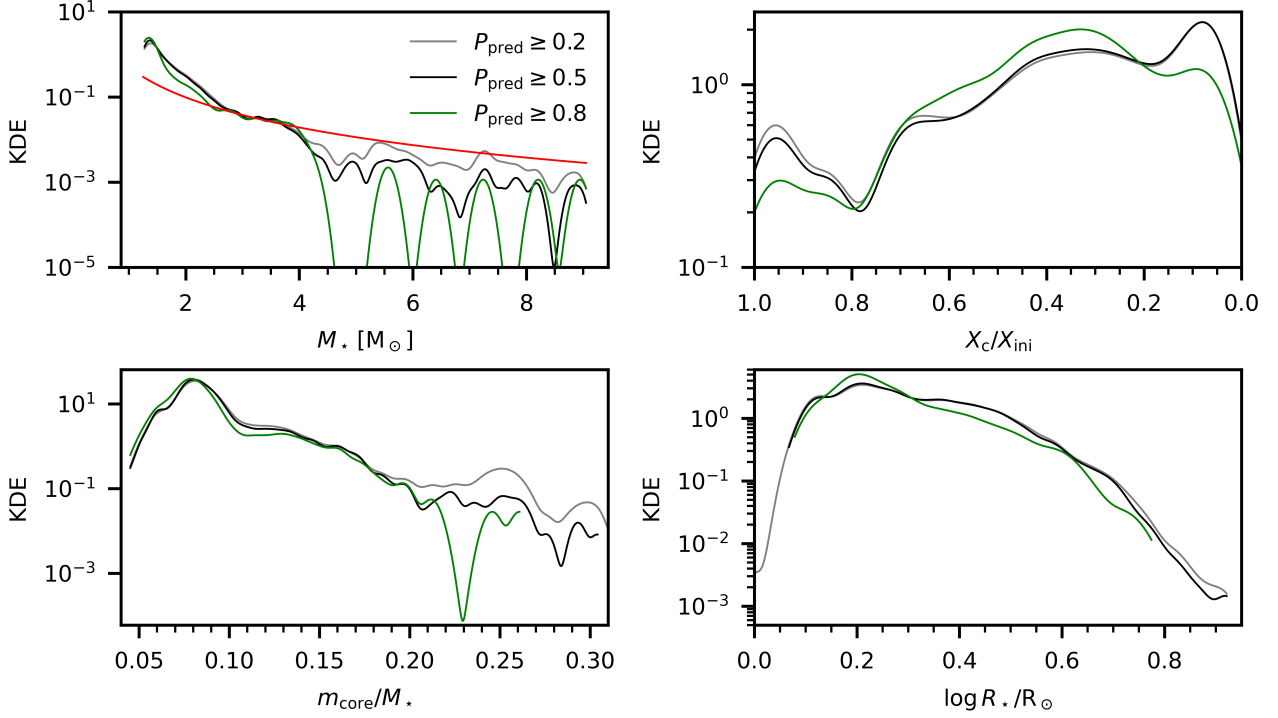


Fig. 3: Kernel density estimations of the MLEs for the stellar mass (top-left), the fraction of initial hydrogen in the core (top-right), the fractional core mass (bottom-left), and the stellar radius (bottom-right). The different colours indicate different threshold on the classification probability from Hey & Aerts (2024) for the star to be considered a g-mode pulsator. The red line in the top-left panel corresponds to a Salpeter IMF with $dN/dM \propto M^{-2.35}$.

Table 2: Summary of p values from two-sample Kolmogorov-Smirnov tests.

Parameter	Sample 1	Sample 2	Z	p value
M_*	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.0045	< 0.001
X_c/X_{ini}	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.0045	0.002
				0.071
m_{cc}/M_*	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.0045	< 0.001
R_*	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.0045	0.477
M_*	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.014	0.0011
X_c/X_{ini}	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.014	0.083
				0.44
m_{cc}/M_*	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.014	< 0.001
R_*	$P_{pred} = 0.2$	$P_{pred} = 0.5$	0.014	0.283
M_*	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.0045	< 0.001
X_c/X_{ini}	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.0045	< 0.001
m_{cc}/M_*	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.0045	< 0.001
R_*	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.0045	< 0.001
M_*	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.014	< 0.001
X_c/X_{ini}	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.014	< 0.001
m_{cc}/M_*	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.014	< 0.001
R_*	$P_{pred} = 0.5$	$P_{pred} = 0.8$	0.014	< 0.001

Notes. The two additional rows for X_c/X_{ini} show the p values when only stars below $4 M_\odot$ are taken into account to avoid under-representation. The p value is based on the null hypothesis that sample 1 and sample 2 are drawn from the same distribution.

grey in Fig. 4. We conclude that the shape of the derived distributions is robust, although the uncertainties at higher masses are larger as a result of the scarcity of higher-mass g-mode pul-

sators. The derived distribution of the stellar masses is compared with a Salpeter initial-mass function (IMF, Salpeter 1955) $dN/dM \propto M^{-2.35}$. We observe an excess of g-mode pulsators below roughly $1.6 M_\odot$. This suggests that the convective flux blocking mechanism is efficient in exciting g modes within a narrow mass range.

Interestingly, we do not observe a drop in the number of g-mode pulsators between the predicted blue edge of the γ Dor instability region (roughly above $2 M_\odot$, Dupret et al. 2005) and the predicted red edge of the SPB instability region (roughly below $3 M_\odot$, Szewczuk & Daszyńska-Daszkiewicz 2017). We perform one-sample Kolmogorov-Smirnov tests assuming a cumulative distribution function predicted by a Salpeter IMF. We adopt as null hypothesis that the sample is drawn from the assumed IMF distribution. Again, we omit stars with a mass above $4 M_\odot$ to avoid under-representation. When using a variable lower limit on the stellar mass and a fixed upper limit of $4 M_\odot$, we find that the mass distribution of our sample of g-mode pulsators is compatible with a Salpeter IMF down to roughly 2.2 - $2.5 M_\odot$, even for the strict threshold of $P_{pred} \geq 0.8$, as shown in Fig. 5. The resulting deduced mass distribution of the *Gaia* pulsators therefore indicates that the κ -mechanism is effective in exciting g modes between the currently assumed disjoint theoretical instability regions of γ Dor and SPB stars.

As already highlighted in the Introduction, observational evidence for pulsators in this part of the Hertzsprung-Russell diagram (HRD) had already been found from high-cadence space photometry assembled with the *Kepler* and TESS space telescopes, prior to the results from *Gaia* DR3 by Gaia Collaboration et al. (2023a) and Aerts et al. (2023), and Hey & Aerts (2024). Even earlier Mowlavi et al. (2013, 2016) detected fast